

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BITTC402 - PROBABILITY AND STATISTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Probability of getting an even number in a throw of a die is
 - a) $1/3$
 - b) $1/6$
 - c) $1/2$
 - d) $2/3$
- 2 What is the chance that a non-leap year contains 53 Sundays?
 - a) $2/7$
 - b) $4/7$
 - c) $3/7$
 - d) $1/7$
- 3 If two unbiased coins are tossed then what is expectation of number Heads
 - a) 0
 - b) 2
 - c) 0.5
 - d) 1
- 4 In binomial Distribution if mean & variance are 12 & 8 respectively then total number of trials 'n' is
 - a) 18
 - b) 24
 - c) 36
 - d) none
- 5 A variable X follows Poisson distribution with mean 3 then $P(X=0)$ is
 - a) 1
 - b) e^{-3}
 - c) e^{-2}
 - d) None
- 6 If a random variable X follows the Poisson distribution such that $P(X=1)=2P(X=2)$ then mean of the distribution is
 - a) 1
 - b) 2
 - c) 3
 - d) None
- 7 If two regression coefficients are 0.8 and 0.6 then correlation coefficient will be
 - a) 0.96
 - b) 0.69
 - c) 0.48
 - d) None
- 8 The Karl Pearson's coefficient of correlation lies between
 - a) +1 and -1
 - b) +2 and -2
 - c) +1/2 and -1/2
 - d) None
- 9 In large sample test, the critical value for 95% confidence level in two tailed test is
 - a) 1.64
 - b) 1.96
 - c) 2.58
 - d) None

10 The critical value in t-test is depends upon

- a) Sample mean b) Population mean c) Degree of freedom and significance level d) None

11 In a Markov chain, the future state depends only on:

- a) All past states b) Only the initial state c) The current state
d) The average of all past states

12 A random variable X has the following probability distribution

X	1	2	3	4
P(X=x)	k	4k	2k	3k

Then the value of k is

- a) 10 b) 0.1 c) 0.5 d) None

Q2. Solve the following.

- A) Three factories A, B, C produce 30 %, 50 % and 20 % of the total production of an item. Out of their production 80 %, 50 % and 10 % are defective. An item is chosen at random and found to be defective. Find the probability that it was produced by the factory A. 6
- B) If four fair coins are tossed . X is the number of Heads that occur, (i) find Probability distribution function of X (ii) Calculate Expectation and Standard Deviation of X. 6

Q3. Solve the following.

- A) A variable X follows Poisson distribution with Variance 3. Find (i) P(X=2) (ii) P(X≥4). 6
- B) The average test marks in a particular class is 79. The standard deviation is 5. If the marks are distributed normally, how many students in class of 200 did not receive marks between 75 and 82? { Given: A(0.7)=0.2580, A(0.8)=0.2881, A(0.6)=0.2257} 6

Q4. Solve Any Two of the following.

- A) Find from the following values of the demand and the corresponding price of commodity, the degree of correlation between the demand and price by computing Karl Pearson's coefficient of correlation : 6

Demands in quintals	65	66	67	67	68	69	70	72
Price in paisa per kg	67	68	65	68	72	72	69	71

- B) Given $6y=5x+90$, $15x=8y+130$, $\sigma_x^2=16$ find (i) mean of x and y (ii) coefficient of correlation 'r' (iii) σ_y^2 6
- C) Ten competitors in a musical test were ranked by 3 judges A, B, C in the following order 6

Rank by A	1	6	5	10	3	2	4	9	7	8
Rank by B	3	5	8	4	7	10	2	1	6	9
Rank by C	6	4	9	8	1	2	3	10	5	7

Using rank correlation method, Discuss which pair of judges has nearest approach to common in music.

Q5. Solve Any Two of the following.

- A) A die was thrown 9000 times and of these 3220 yielded a 3 or 4, can the die be regarded as unbiased? 6
- B) A soap manufacturing company was distributing particular brand soap through a large number of retail shops. Before a heavy advertisement campaign, the mean sale per week per shop was 140 dozens. After campaign a sample of 26 shops was taken the mean of sale was found to be 147 dozens with S.D. of 16. Can you consider advertisement effective? Use level of significance 5% { 6

Given $t_{0.05}=2.06$ when $DOF=25$

- C) The following table gives the number of accidents in a city during a week. Use Chi Square test to find whether the accidents are uniformly distributed over a week. 6

Day	MON	TUE	WED	THU	FRI	SAT	SUN	TOTAL
No. of accidents	13	15	9	11	12	10	14	84

{ Given: .Table value of Chi square at 6 d.o.f. when LOS 5% is 12.59 }

Q6. Solve Any Two of the following.

- A) Define Markov process and Stochastic process. Explain transition probability Matrix and its diagrammatic representation. 6
- B) Using least square method, fit the straight line $y=a+bx$ to the following data 6

X	0	5	10	15	20	25
Y	12	15	17	22	24	30

- C) Fit a second degree curve to the following data 6

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

*** End ***